

Expressive Power of Languages

- When the domain relational calculus is restricted to safe expressions, it is equivalent in expressive power to the tuple relational calculus restricted to safe expressions.
- All three of the following are equivalent:
 - The basic relational algebra (without the extended relational-algebra operations)
 - The tuple relational calculus restricted to safe expressions
 - The domain relational calculus restricted to safe expressions
- Domain relational calculus also does not have any equivalent of the aggregate operation. It can be extended to support aggregation, and extending it to handle arithmetic expressions is straightforward.



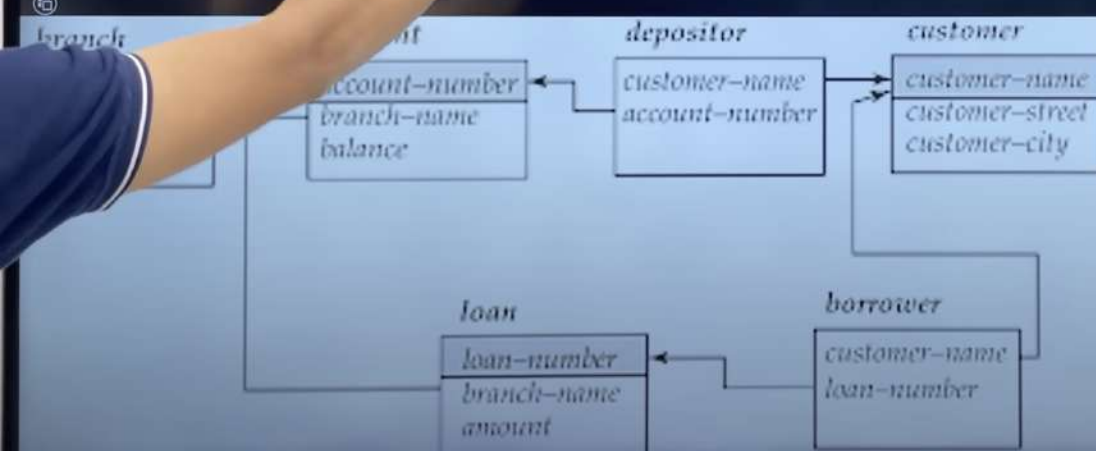
Safety of Expressions

- Similar to TRC we can have an unsafe expression that may generate infinite relations.
- An expression such as
 - $\{ \langle i, n, d, s \rangle \mid \neg (\langle i, n, d, s \rangle \in \text{instructor}) \}$
- is unsafe, because it allows values in the result that are not in the domain of the expression.



Q Find the loan number for each loan of amount over 1200?

$\{ \langle l \rangle \mid \exists_{b,a} \langle l, b, a \rangle \in \text{loan} \wedge a > 1200 \}$



Student(Roll No, Name, Branch)

Q Find the Roll No of all computer science students?

SQL: select Roll No from student where branch = CSE

RA: $\{\pi_{\text{Rname}} (\sigma_{\text{branch} = \text{CSE}} (\text{Student}))\}$

TRC: $\{t. \text{Roll No} \mid \text{Student}(t) \wedge t. \text{branch} = \text{CSE}\}$

DRC: $\{(\text{Roll No}) \mid \text{Student} (\text{Roll no, Name, Branch}) (\text{Name, Branch}) \wedge \text{branch} = \text{CSE} \}$

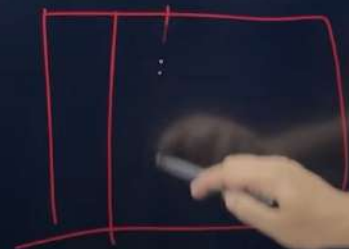
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Domain Relational Calculus

- Domain calculus differs from the tuple calculus in the type of variables used in formulas: rather than having variables range over tuples.
- The variable range over single values from domains of attributes. To form a relation of degree n for a query result, we must have n of these domain variables. One for each attribute.
- An expression of the domain calculus is of the form $(x_1, x_2, \dots, x_n \mid \text{COND}(x_1, x_2, \dots, x_n, x_{n+1}, x_{n+2}, \dots, x_{n+m}))$
- Where $x_1, x_2, \dots, x_n$ are domain variables that range over domains and COND is a condition of the domain relational calculus.



Chapter-8 (Relational Calculus)

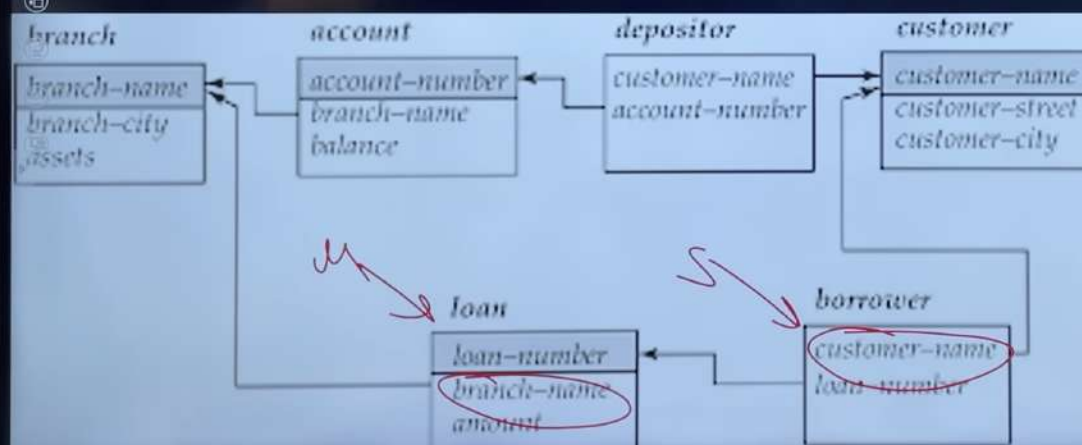
Tuple Relation Calculus

Q Find the name of all the customers who have a loan from Noida branch?

$\{t \mid \exists s \in \text{borrower } (t[\text{customer name}] = s[\text{customer name}])$

$\wedge \exists u \in \text{loan } (u[\text{customer name}] = s[\text{customer name}])$

$\wedge u[\text{branch name}] = \text{'Noida'} \}$



Chapter-8 (Relational Calculus)

Student(Roll No, Name, Branch)

Tuple Relation Calculus

Q Find the Roll No of all computer science students?

SQL: select Roll No from student where branch = CSE

RA: $\{\pi_{\text{roll no}}(\sigma_{\text{branch} = \text{CSE}}(\text{Student}))\}$

TRC: $\{t. \text{Roll No} \mid \text{Student}(t) \wedge t. \text{branch} = \text{CSE}\}$

DRC:



Student(Roll No, Name, Branch)**Q Find the details of all computer science students?**

SQL: select * from student where branch = CSE

RA: $\{\sigma_{\text{branch} = \text{CSE}}(\text{Student})\}$ TRC: $\{t \mid \text{Student}(t) \wedge t.\text{branch} = \text{CSE}\}$

DRC:

...

KG

Tuple Relational Calculus

- TRC is based on specifying a number of tuple variables. Each tuple variable usually range over a particular database relation, meaning that the variable may takes as its value from any individual tuple of a relation.
- A simple tuple relational calculus query is of the form.
 - $\{t \mid \text{Condition}(t)\}$
- Where t is a tuple variable and $\text{condition}(t)$ is a conditional expression involving. The result of such a query is the set of all tuple t that satisfy condition (t) .
- We use $t[A]$ or $t.A$ to denote the value of tuple t on attribute A . We use $t \in r$ or $r(t)$ to denote that tuple t is in relation r .



4:09:32 / 5:33:30 • (Chapter-8: Relational Calculus)- Overview, Tuple Relation Calculus, Domain Relati...



Relational Calculus

- Relational calculus is non-procedural query language, where we have to define what to get and not how to get it

